



Occurrence of emerging sulfonamide resistance (*sul1* and *sul2*) associated with mobile integrons-integrase (*int11* and *int12*) in riverine systems

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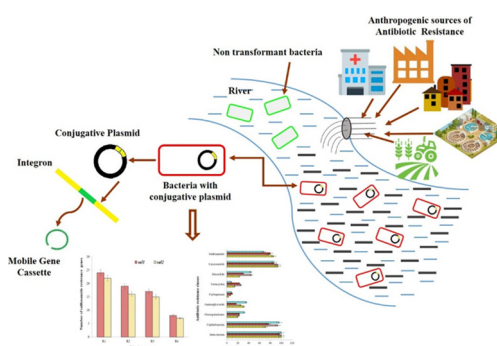
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HIGHLIGHTS

- *Int11* (60.33%) and *int12* (9.09%) were found from different geo-climatic zones.
- Integrons were associated with *sul1* (85.18%) and *sul2* (76.54%) genes in rivers.
- *blaTEM* (93.0%) was the most prevalent ESBL gene in riverine system.
- MDR was observed for macrolides, aminoglycosides and fluoroquinolones in aquifers.
- Frequencies of *Int11*, *sul1*, and *sul2* genes were higher in studied riverine system.

GRAPHICAL ABSTRACT



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ABSTRACT

Global use of antibiotics has exceedingly enhanced in agricultural, veterinary and prophylactic human use in recent days. Hence, these antibiotics can easily be found in the environment. This study revealed the occurrence of emerging MDR and ESBL producing strains, pollution profile, and factors integrons (*int11* and *int12*) and environmental factors associated, in the riverine systems under different ecological and geo-climatic zones were investigated. The samples were collected based on anthropogenic intervention such as discharge of domestic wastes, industrial wastes, hospital, and municipal wastes. Among 160 bacterial morphotypes, 121 (75.62%) exhibited MDR trait with maximum resistance towards lincosamide (CD = 71.3%), beta-lactams (P = 70.6%; AMX = 66.3%), cephalosporin (CZ = 60.6%; CXM = 34.4%), sulfonamide (COT = 50.6%; TR = 43.8%) followed by macrolide (E = 29.4%), tetracycline (TET = 18.8%), aminoglycosides (S = 18.8%; GEN = 6.3%), fluoroquinolones (NX = 18.1%; OF = 4.4%) and carbapenem (IPM = 5.0%). *Int11* gene was detected in 73 (60.3%) of isolates, whereas *int12* was found in 11 (9.09%) isolates. Eight (6.61%) isolates carried both integron genes (*int11* and *int12*). *sul1* and *dhfrA1* genes were detected in 53 (72.6%) and 63 (86.3%) isolates, respectively. A total of 103 (85.1%) were found ESBL positive with the presence of ESBL genes in 100 (97.08%) isolates. In riverine systems most prevalent ESBL gene *blaTEM* (93.0%) was detected alone as well as in combination with *bla* genes. The

Abbreviations: BOD, biochemical oxygen demand; COD, chemical oxygen demand; TDS, total dissolved solid; TSS, total suspended solid; PCR, polymerase chain reaction; CFU, colony-forming unit; MDR, multidrug resistance; ARB, antibiotic resistance bacteria; ARGs, antibiotic resistance genes; AST, antibiotic sensitivity testing; MAR, multiple antibiotic resistance; AMR, antimicrobial resistance.

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data can be utilized for public awareness and regulation of guidelines by local governing bodies as an alarming threat to look-out against the prevalent resistance in environment thereby assisting in risk management during epidemics. This study is a comprehensive investigation of emerging antibiotic pollutants and its resistance in bacteria associated with factors integrons-integrase responsible for its dissemination. It may also assist in global surveillance of antibiotic resistance and policies to curtail unnecessary antibiotic use.

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1. Introduction

The propagation of emerging antibiotic resistance has significantly affected the environment including humans and animals. The discharges of effluents in aquatic environment are main cause for development of antibiotics resistance in bacteria through selection pressure as well as uptakes of antibiotics by aquatic organisms and crops irrigated with polluted water, paving its way to entering in food chain and human gut. The reason behind “the global resistome” or antimicrobial resistance (AMR) including excessive use and misuse of antibiotics in humans and animals (Aslam et al., 2018). Antimicrobial resistance is an increasingly global threat to public health, which requires immediate action across all sectors of society. The global consumption of antimicrobials in livestock has indicated hotspot of use of antibiotic across continents that will have an impact on economic and public health in the coming years (vanBoeckel et al., 2015; Aslam et al., 2018). The pace of discovery of new antibiotics is far left behind to overcome the resistance threat alone. The major factor responsible for this soon-to-be-pandemic outbreak includes excessive and irrational use and sale of antibiotics in hospitals, industries, animal farms, veterinary, domestic (Kotlarska et al., 2015), and agriculture sectors (Hamelin et al., 2007; Nielsen et al., 2018).

Wastewater released from these sources is loaded with high concentrations of residual antibiotics as these do not get completely metabolized upon consumption and enter the natural habitats and aquatic environments (Heuer et al., 2011). India is one of the leading producers of pharmaceuticals in the world (Rehman et al., 2015) and hence the rivers are prone to become a reservoir of emerging contaminants antibiotic and its pollution as a consequence of uninterrupted discharges from pharmaceutical industries and hospitals. Rivers are polluted with industrial and domestic wastewater, which contains nutrient rich environment and other favorable condition to become hotspot for antibiotic resistance. Such aquatic reservoirs favor the vertical and horizontal gene transfer, which significantly involve in dissemination of ARGs (Hultman et al., 2018; Sivagami et al., 2020). Effluents generated from these pharma industries are known to contain high levels of residues of drugs belonging to classes - fluoroquinolones, sulfonamides and beta-lactams (Fick et al., 2009). High concentration of sulfonamide in environment may be attributed to its large scale application in treating common bacterial infections and its low cost (Suzuki and Hoa, 2012). Irrespective of its nature and origin i.e., glacier, plain, plateau or mountain; the rivers are exposed to anthropogenic intervention subsequently leading to pollution due to the vast numbers of prevailing pharmaceutical industries. Further religious activities like mass bathing and gatherings on river banks during festivals also serve as potential source of dissemination of resistant bacteria (Purohit et al., 2020).

These antibiotics which exist in ng/L to µg/L in wastewater and their prolonged presence in environment can cause the emergence of highly resistant and transformed antibiotic resistant genes and bacterial strains (Li and Zhang, 2010). Discharge of such contaminated effluents into the receiving water bodies (rivers/lakes) turns the latter into rich hotspots of bacteria harbouring genes responsible for antibiotic resistance and its dissemination. Such environments rich in organic nutrients provide suitable platforms for the lateral shift of resistant genes by mobile genetic elements (mainly integrons) to autochthonous bacteria (Saswati et al., 2013; Chow et al., 2015). Integrons serve as vehicles for the easy dissemination of resistant genes in open environments (Yang et al.,

2014). These DNA fragments (Makowska et al., 2020) are located on plasmids, chromosomes, and transposons (Berhane and Meniyamer, 2015) and are equipped to acquire, capture, mobilize, integrate and express genes encoding antibiotic resistance (Nnadozie and Odume, 2019; Peterson and Kaur, 2018). Generally, integron consists of two conserved segments one each in 5' and 3' regions flanked by a variable region (Adesoji et al., 2017; Canal et al., 2016). The 5'CS contains an *intI* gene that encodes integrase enzyme, a recombination site (*attI*), a common promoter, whereas the 3' CS consists of *qacEΔ1* gene that confers resistance to quaternary ammonium compounds, a *sul1* gene for sulfonamide resistance and an Open Reading Frame (ORF) region (*orf5*) (Sáenz et al., 2010). The variable region consists of a combination of gene cassettes that encodes genes for antibiotic resistance. The promoter region is responsible for the expression of the captured gene cassettes (Sáenz et al., 2010; Adesoji et al., 2017). Gene cassettes are captured via recombination between the *attI* site of the integron and *attC* site from the cassette (Kotlarska et al., 2015).

Integron-integrase genes (*intI*) are control the activity of integron. At the integron recombination site (*attI*), *intI* catalyses the insertion of exogenous gene cassettes (Ghaly et al., 2019). Integrons are best known for their role in assisting to facilitate the global antibiotic-resistance crisis. Their capture and expression of resistance genes has allowed integron-carrying cells to rapidly respond to antibiotic selection pressures (Palme et al., 2017). All together these segments make up the integrons, known for its ability to import different combinations of antibiotic resistance genes among various non-pathogenic strains thereby turning them into MDR bacteria (Cambray et al., 2010). This property of the integron renders it to be a proxy element for representing the antibiotic pollution load in an environment (Gillings, 2017; Amos et al., 2018). Among the three classes of integrons, class 1 integron is the most widely disseminated genetic element responsible for the spread of antibiotic resistance among bacteria (Cambray et al., 2010).

Thus, the present research was conducted to assess and analyze the extent of antibiotic resistance threat in Indian perennial aquatic reservoirs from different geo-climatic and ecological zones: Ganga, Gomti, Narmada and Kaveri. The study explores the occurrence of emerging sulfonamide resistance (*sul1* and *sul2*) and its association with mobile integrons-integrase (*intI1* and *intI2*). In addition, it also contains diversified susceptibility/resistance pattern of rivers towards β-lactam and thirteen other antibiotics groups. The study is focused on the detection of ARB in surface water and characterization of the integrons responsible for the resistant traits thereby providing evidence regarding the latter as repositories of ARB and ARGs. The presence of ARB pheno- and geno-types in the river may have several adverse impacts on the aquatic ecosystem. Hence, this type of finding is alarming and needs stringent surveillance for setting up of urgent measures to overcome and prevent the spread of antibiotic resistance.

2. Material and methods

2.1. Study site and sample collection

Water samples were collected from four different ecological conditions and geo-climatic zones comprising riverine systems in India. In this study, the rivers were selected from northern, central and southern zones of the country comprising Himalaya glacial habitat; central hilly region; and Southern plateau. The climatic conditions varied hugely

with optimum temperatures ranging from 4 °C (Ganga) to 40 °C (Kaveri) in these zones. Moreover the studied rivers hold huge socio-economic and cultural importance in the country. The river Ganga, ~2525 km long is a trans-boundary river of glacial origin; Gomti, a plain-fed river travelling ~960 km is its tributary; Narmada, a river in central India ~1312 km long originates from Amarkantak plateau; Kaveri, ~805 km long rising from Western Ghats and flows through the southern states of India. At each location, five sites having anthropogenic intervention such as discharge of domestic wastes, industrial wastes, hospital, and municipal wastes were selected for sample collection. The samples were collected from June 2019 to December 2019. The water samples were collected by grab sampling method from ~15 cm below the surface from midstream of the river. The collected samples were stored in sterilized glass bottles at 4 °C till further processing.

2.2. Analysis of physico-chemical parameters and metals

The samples were tested against an exhaustive list of physico-chemical parameters to analyze the water quality of the river systems. The analysis was performed as per BIS (2012) and APHA methods (2017). Test for parameters such as pH, Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Dissolved Solids (TDS), Total Suspended Solid (TSS), alkalinity, phosphate, sulfate, and nitrate were conducted in the laboratory as per APHA and BIS methods 2012 whereas, temperature, EC and Dissolved Oxygen (DO) was measured on-site using a calibrated multi-parameter meter U-52 (Horiba, Japan). Analysis of metals was performed using Atomic Absorption Spectroscopy (AAS) (Pin AAcle 990F, Perkin Elmer, Waltham, Massachusetts, U.S.) instrument for metals such as Cd, Cr, Zn, Pb, Ni, Mn, Ca, and Mg. The concentration of metals is expressed in mg/L.

2.3. Isolation of bacterial cultures

Culturable heterotrophic bacteria were isolated from the water samples using the spread plate method as per the APHA method no. 9125 C (2017). For quantitative analysis, a series of logarithmic dilutions (10^{-1} to 10^{-4}) were prepared in sterile saline solution (0.9% NaCl). Each dilution was inoculated into Plate Count Agar (HiMedia, India). Plates were incubated at 37 °C for 24–26 h. Following incubation, colonies on each plate were counted and isolated based on their morphological features. Pure cultures of all the isolates were subjected to Gram's reaction.

2.4. AST and MAR indexing of isolates

The sensitivity of bacterial isolates towards various antibiotics was assessed using the disk diffusion method given by Kirby-Bauer (CLSI, 2017). A total of 14 antibiotics belonging to 9 different antimicrobial groups were used: Beta-lactam-penicillin (P; 30 µg), amoxicillin (AMX; 30 µg); cephalosporin-cefazoline (CZ; 30 µg), cefuroxime (CXM; 30 µg); fluoroquinolones-norfloxacin (NX; 10 µg), ofloxacin (OF; 5 µg); aminoglycosides-streptomycin (S; 10 µg), gentamicin (GEN; 10 µg); carbapenems-imipenem (IPM; 750 µg), tetracyclin-tetracyclin (TE; 30 µg); macrolides-erythromycin (E; 15 µg); lincosamides-clindamycin (CD; 2 µg); sulfonamides-co-trimoxazole (COT; 25 µg), trimethoprim (TR; 25 µg). The results of inhibitory zones were interpreted according to Clinical Laboratory Standards Institute (CLSI) specifications (CLSI, 2017). Isolates that displayed resistance to three or more different classes of antibiotics were classified as Multidrug-resistant bacteria (Smyth et al., 2020). The antibiotic-resistant frequency was calculated as: Antibiotic-resistant frequency (%) = $(a/b) \times 100$ (Su et al., 2012) where 'a' is indicating the number of isolates resistant to antibiotics and 'b' is the no. of isolated from the sample. The antibiotic disks were commercially obtained from HiMedia Laboratories Pvt. Ltd. India. Based on the resistance phenotype obtained, the MAR index of the isolates was calculated following Krumperman (1983).

2.5. Screening of extended-spectrum β -lactamases (ESBLs)

The MDR isolates were phenotypically tested for the production of beta-lactamase enzymes such as ESBL by double-disk synergy test (DDST) method using ceftazidime (CAZ 30 µg), cefotaxime (CTX 30 µg) and an augment in- AMC (20 µg amoxicillin and 10 µg clavulanic acid) disk to check for the presence of ESBL production (CLSI, 2013). An increase in the inhibition zone of the cephalosporin disk in the direction of the augment in disk indicates a positive result.

2.6. Detection of integrons

Genomic DNA was extracted from bacterial cultures positive for MDR phenotype by the alkaline lysis method. The purity and concentration of genomic DNA were analyzed by NanoDrop spectrophotometer (ND-1000, Thermo Scientific, US) and agarose gel electrophoresis (Makowska et al., 2020). The presence of integrase gene *int1* and *int2* genes in MDR positive isolates was determined by PCR (Table 1). The reaction conditions were described as initial denaturation at 94 °C for 5 min followed by 30 cycles of denaturation at 94 °C for the 30 s, annealing at 55.5 °C for 30 s and extension at 72 °C for 1 min; and a final elongation step at 72 °C for 5 min (Mazel et al., 2000). The amplified products were visualized under UV transilluminator (G: BOX Syngene, USA) after agarose gel electrophoresis. A 100-bp and 1-kb DNA ladder (New England Biolabs, USA) was used as a marker.

2.7. Detection of resistance determinants located on integrons and beta-lactamase genes

The presence of sulfonamide (co-trimoxazole and trimethoprim) determinants (*sul1*, *sul2*, and *dfrA1*, *dfrA12*) and tetracycline determinants was analyzed by PCR (Eppendorf AG, Germany). The thermocycling conditions were as follows: initial denaturation at 95 °C for 3 min followed by 35 cycles of denaturation at 94 °C for the 30 s, annealing at 55 °C for 30 s and extension at 72 °C for 1 min; and a final elongation step at 72 °C for 10 min. In addition, tetracycline thermocycling conditions were performed for initial denaturation (95 °C, 3 min) followed by 35 cycles of denaturation (94 °C, 30 s), annealing (56 °C, 30 s), extension (72 °C, 1 min); and a final elongation step (72 °C, 10 min) (Maynard et al., 2003). Multiplex PCR was performed to detect *blaTEM*, *blaSHV*, *blaOXA* and *blaCTX-M* (CTX-M1, CTX-M2, and CTX-M9) genes using the conditions as described: initial denaturation at 94 °C for 10 min followed by 30 cycles of denaturation at 94 °C for the 40 s, annealing at 60 °C for 40 s and extension at 72 °C for 1 min; and a final elongation step at 72 °C for 7 min (Dallenne et al., 2010). PCR products were confirmed by gel electrophoresis (1% agarose) and visualized under UV transilluminator for the desired band size using 100-bp and 1-kb DNA ladder (New England Biolabs, USA) as a marker (Table 1).

2.8. Statistical analysis

Data for the water quality analysis is expressed as the mean $\pm$ SD. Correlations among physico-chemical parameters, sulfonamide resistance genes, and integrons were determined with Pearson's correlation coefficient.

3. Result and discussion

3.1. Water quality and metal analysis

The water samples collected from different rivers were tested for various physico-chemical parameters such as pH, DO, EC, BOD, COD, TSS, TDS, alkalinity, sulfate (SO_4^{2-}), phosphate (PO_4^{3-}), nitrate (NO_3^-) and heavy metals (Table 2). The water quality of rivers (Ganga, Gomti, Narmada and Kaveri) was found to be affected considerably. The values of some physico-chemical parameters such as BOD, TSS, NO_3^- , Cd, Mn

Table 1
Details of the primers used in this study.

Primer name	Target gene	Primer sequence (5'-3')	Amplicon size (bp)	References
Primers for genes				
SulI-F	sul1	F-TTC GGC ATT CTG AAT CTCAC	822	Mazel et al., 2000
SulI-R		R-ATG ATC TAA CCC TCG GTC TC		
SulII-F	sul2	F-CGG CAT CGT CAA CAT AAC C	722	Mazel et al., 2000
SulII-R		R-GTG TGC GGA TGA AGT CAG		
Tet(A)-F	tet(A)	F-GTG AAA CCC AAC ATA CCC C	880	Maynard et al., 2003
Tet(A)-R		R-GAA GGC AAG CAG GAT GTA G		
DfrA1-F	dfrA1	F-AAG AAT GGA GTT ATC GGG AAT G	391	Maynard et al., 2003
DfrA1-R		R-GGG TAA AAA CTG GCC TAA AAT TG		
DfrA12-F	dfrA12	F-GGTGSGCAGAAGATTTTCGC	319	Navia et al., 2003
DfrA12-R		R-TGGGAAAGAAGGCGTCACCTC		
IntI1-F	intI1	F-GGG TCA AGG ATC TGG ATT TCG	483	Maynard et al., 2003
IntI1-R		R-ACA TGG GTG TAA ATC ATC GTC		
IntI2-F	intI2	F-CAC GGA TAT GCG ACA AAA AGG T	788	Maynard et al., 2003
IntI2-R		R-GTA GCA AAC GAG TGA CGA AAT G		
Primers for ESBL genes				
TEM (F)	blaTEM	F-CAT TTC CGT GTC GCC CTT ATT C	800	Dallenne et al., 2010
TEM (R)		R-CGT TCA TCC ATA GTT GCC TGA C		
SHV (F)	blaSHV	F-AGC CGC TTG AGC AAA TTA AAC	713	
SHV (R)		R-ATC CCG CAG ATA AAT CAC CAC		
OXA (F)	blaOXA	F-GGC ACC AGA TTC AAC TTT CAA G	564	
OXA (R)		R-GAC CCC AAG TTT CCT GTA AGTG		
CTX-M1(F)	blaCTX-M1	F-TTA GGA ART GTG CCG CTG YA	688	
CTX-M1(R)		R-CGA TAT CGT TGG TGG TRC CAT		
CTX-M2(F)	blaCTX-M2	F-CGT TAA CGG CAC GAT GAC	404	
CTX-M2(R)		R-CGA TAT CGT TGG TGG TRC CAT		
CTX-M9(F)	blaCTX-M9	F-TCA AGC CTG CCG ATC TGG T	561	
CTX-M9(R)		R-TGA TTC TCG CCG CTG AAG		

exceeded the limits prescribed by the national standard authorities (BIS, 2012). In recent study by Boger et al. (2020) they reported that Belem river Brazil, pH, COD, BOD and temperature were recorded as 6.0, 34 mg/L, 4 mg/L, and 22.5, respectively whereas, COD and BOD of Barigui river were found 15 mg/L and 3 mg/L, respectively. This may be due to the seasonal-temporal variations and challenges such as unnecessary encroachment, unregulated pollution, and agricultural flow faced by the river. The concentration of most of the metals was found within the permissible limits indicating no evidence of metal pollution (Table 2).

3.2. Enumeration of culturable bacterial strains

Isolation of bacterial colonies was done by the spread plate method. The isolates were categorized based on biochemical, physiological, and morphological characters. The overall bacterial colonies count varied in each sample. The heterotrophic bacterial counts ranged from 2.3×10^5 to 5.7×10^6 CFU/mL in Ganga, 1.9×10^5 to 4.2×10^6 CFU/mL in Gomti,

1.2×10^5 to 3.9×10^6 CFU/mL in Narmada and 2.1×10^5 to 3.5×10^6 CFU/mL in Kaveri (Table 3). The bacterial count differed at each sampling location depending upon the magnitude of prevailing anthropogenic activities. Among 160 total bacterial isolates, 114 (71.25%) isolates were found to be Gram-negative and 46 (28.75%) were Gram-positive. In a similar type study, the heterotrophic bacterial count ranged from 7.4×10^2 to 1.9×10^3 CFU/mL in upstream and 1.7×10^3 to 2.9×10^4 CFU/mL in downstream (Makowska et al., 2016) 3.0×10^2 to 1.7×10^3 CFU/mL in the upstream river, and 6.8×10^2 to 3.0×10^3 CFU/mL in the downstream river (Koczura et al., 2012) and 0.6×10^5 to 2.2×10^5 CFU/mL in river water (Obayiwana et al., 2018).

3.3. Susceptibility screening and MAR index of isolates

The isolates were subjected to multidrug resistance (MDR) screening to 14 antibiotics belonging to 9different classes (beta-lactam, cephalosporin, fluoroquinolones, aminoglycosides, carbapenems tetracycline, macrolides, lincosamides, and sulfonamides). A total of 121 (75.62%)

Table 2
Water quality analysis (Physico-chemical and metals) of the studied rivers.

Physico-chemical parameters													
River	pH	DO (mg/L)	Temp (°C)	EC (µS/cm)	BOD (mg/L)	COD (mg/L)	TSS (mg/L)	TDS (mg/L)	Alkalinity (mg/L)	SO ₄ ²⁻ (mg/L)	PO ₄ ³⁻ (mg/L)	NO ₃ ⁻ (mg/L)	
R1	7.77 ± 0.20	5.28 ± 0.04	28 ± 1.0	334 ± 1.0	14 ± 2.83	54 ± 0.31	56 ± 2.00	232 ± 1.15	24 ± 1.00	31.26 ± 0.54	0.55 ± 0.04	0.15 ± 0.01	
R2	7.82 ± 0.15	4.40 ± 0.19	29 ± 0.15	465 ± 1.0	35 ± 4.24	134 ± 2.00	246 ± 2.00	256 ± 2.00	308 ± 1.00	23.33 ± 0.10	0.53 ± 0.12	0.26 ± 0.01	
R3	7.28 ± 0.02	6.07 ± 0.03	27 ± 0.21	311 ± 1.0	26 ± 2.83	106 ± 0.20	80 ± 2.00	208 ± 2.00	59 ± 1.00	20.50 ± 0.10	0.43 ± 0.01	0.13 ± 0.02	
R4	7.51 ± 0.02	5.80 ± 0.10	26 ± 0.06	252 ± 1.00	26 ± 2.83	98.4 ± 0.40	408 ± 2.00	138 ± 2.00	203 ± 0.58	25.36 ± 0.55	0.75 ± 0.01	0.48 ± 0.01	
Metals analysis													
River	Cd (mg/L)	Cr (mg/L)	Pb (mg/L)	Zn (mg/L)	Ni (mg/L)	Ca (mg/L)	Mg (mg/L)	Mn (mg/L)					
R1	BDL	BDL	BDL	0.0035 ± 0.0	BDL	37.0 ± 0.47	3.322 ± 0.01	0.0110 ± 0.01					
R2	0.002 ± 0.0	0.333 ± 0.0	0.001 ± 0.0	0.0470 ± 0.0	0.002 ± 0.0	49.4 ± 0.20	12.537 ± 0.01	0.0100 ± 0.00					
R3	0.005 ± 0.0	BDL	0.005 ± 0.0	BDL	BDL	41.7 ± 0.06	3.260 ± 0.06	0.0400 ± 0.00					
R4	0.002 ± 0.0	BDL	0.001 ± 0.0	0.5800 ± 0.0	0.001 ± 0.0	41.0 ± 1.00	11.658 ± 0.05	0.0140 ± 0.00					

R1: Ganga river; R2: Gomti river; R3: Narmada river; R4: Kaveri river.
BDL: below detection limit; EC: electric conductivity; Cd: cadmium; Cr: chromium; Pb: lead; Zn: zinc; Ni: nickel; Ca: calcium; Mg: magnesium; Mn: manganese.

Table 3
MAR index and CFU count of different sites of riverine system.

River	Sampling sites	MAR index	CFU/mL
Ganga	Site 1	0.45 ± 0.01	2.5 × 10 ⁵
	Site 2	0.43 ± 0.01	2.3 × 10 ⁵
	Site 3	0.42 ± 0.02	3.1 × 10 ⁵
	Site 4	0.41 ± 0.01	3.8 × 10 ⁶
	Site 5	0.53 ± 0.04	5.7 × 10 ⁶
Gomti	Site 6	0.46 ± 0.02	1.9 × 10 ⁵
	Site 7	0.54 ± 0.01	2.4 × 10 ⁵
	Site 8	0.49 ± 0.01	2.8 × 10 ⁶
	Site 9	0.44 ± 0.02	2.5 × 10 ⁵
	Site 10	0.49 ± 0.02	4.2 × 10 ⁶
Narmada	Site 11	0.38 ± 0.02	1.2 × 10 ⁵
	Site 12	0.42 ± 0.03	2.3 × 10 ⁵
	Site 13	0.44 ± 0.02	2.8 × 10 ⁵
	Site 14	0.51 ± 0.03	3.1 × 10 ⁶
	Site 15	0.58 ± 0.02	3.9 × 10 ⁶
Kaveri	Site 16	0.37 ± 0.01	2.1 × 10 ⁵
	Site 17	0.53 ± 0.02	2.8 × 10 ⁵
	Site 18	0.73 ± 0.02	3.6 × 10 ⁵
	Site 19	0.49 ± 0.03	3.5 × 10 ⁶
	Site 20	0.45 ± 0.01	3.8 × 10 ⁵

isolates displayed the MDR trait i.e., was resistant to three or more antibiotics of different classes. Worldwide, isolates from similar aquatic sources have been reported with a high percentage of resistance characteristic: 87.5% in South China (Su et al., 2012), 65.8% in Nigeria (Obayiwana et al., 2018), 86.6% in India (Kaushik et al., 2018) and 75.8% in Ethiopian rivers (Berhane and Meniyamer, 2015).

The highest antibiotic resistance frequency was recorded against antibiotics belonging to lincosamide (CD = 71.3%), beta-lactam group

(P = 70.6%; AMX = 66.3%), cephalosporin (CZ = 60.6%; CXM = 34.4%), sulfonamide (COT = 50.6%; TR = 43.8%) followed by macrolide (E = 29.4%), tetracycline (TET = 18.8%), aminoglycosides (S = 18.8%; GEN = 6.3%), fluoroquinolones (NX = 18.1%; OF = 4.4%) and carbapenem (IPM = 5.0%) (Fig. 1, Table 4).

In another study resistance frequency of trimethoprim (TR = 41.7%), tetracycline (TET = 25%) and imipenem (IPM = 16.7%) were reported in Funan river china (Tuo et al., 2018). The MAR index of maximum isolates were found to be >0.2, thus reflecting high risk of infection by the resistant isolates and greater degree of antibiotic pollution (Table 3). Isolates resistant to beta-lactams and sulfonamide group has been reported in distinct aquatic environments globally (Su et al., 2012; Sabri et al., 2018; Obayiwana et al., 2018) thus supporting high usage and overall development of resistance towards these most commonly used antibiotics. Beta-lactams, cephalosporins, sulfonamides, tetracyclines, aminoglycosides, and fluoroquinolones are the popular antibiotic groups against which resistant studies in riverine sources have yielded positive results (Obayiwana et al., 2018; Koczura et al., 2012; Kaushik et al., 2018; Makowska et al., 2020; Farkas et al., 2016; Chen et al., 2010). The pace of unregulated anthropogenic activities combined with the unnecessary usage of antibiotics may be the primary factor responsible for the expansion of resistant domain.

3.4. Detection of integrons (*int1* and *int2*) by PCR

A total of 121 (75.62%) MDR positive isolates were screened for the detection of integrase genes *int1* and *int2* encoding class 1 and class 2 integron-integrase gene using PCR. *Int1* gene was detected in 73 (60.33%) isolates, whereas *int2* was found in 11 (9.09%) isolates. A total of 8 (6.61%) isolates carried both integrase genes (*int1* and *int2*)

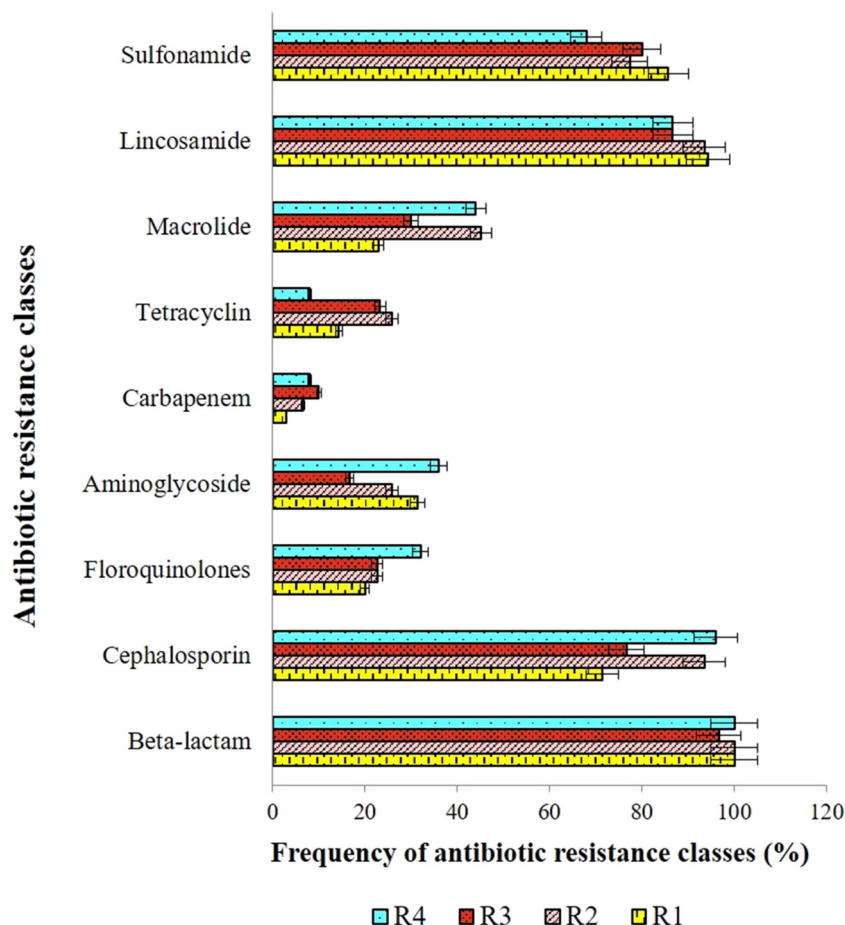


Fig. 1. Distribution of antibiotic resistant classes in (R1-Ganga, R2-Gomti, R3-Narmda, and R4-Kaveri) riverine systems.

Table 4
Antibiotic resistance profiling and distribution of integrase genes in MDR positive isolates.

S. no	Isolate code	Antibiotic phenotypic resistance	ESBL phenotype	Antibiotic genes detected	Beta-lactam genes	Integrans
1.	R1-1	AMX, P, CZ, CXM, GEN, S, IPM, COT	+	<i>sul1, sul2</i>	<i>blaTEM</i> + <i>blaSHV</i>	<i>int1</i>
2.	R1-2	AMX, P, CZ, CXM, CD, COT	+		<i>blaTEM</i> + <i>blaSHV</i>	<i>int1, int2</i>
3.	R1-3	AMX, P, CZ, CXM, S,TET,CD,TR	+	<i>dfrA1, tet(A)</i>	<i>blaTEM</i> + <i>blaSHV</i>	<i>int1</i>
4.	R1-4	AMX, P, GEN, CD, TR	+	<i>dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
5.	R1-5	AMX, NX, CD,COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaCTX-M1</i>	<i>int1</i>
6.	R1-6	AMX, P, CZ, CD, COT, TR	+	<i>sul1, sul2, dfrA1, dfrA12</i>	<i>blaTEM</i> + <i>blaCTX-M1</i> + <i>blaCTX-M9</i>	<i>int1, int2</i>
7.	R1-7	AMX, P, CZ, CD,COT	+	<i>sul1, sul2</i>	<i>blaTEM</i>	<i>int1</i>
8.	R1-8	AMX, P, CZ, CD	+		<i>blaTEM</i>	
9.	R1-9	AMX, P, CZ, CD, COT	+	<i>sul1, sul2</i>	<i>blaTEM</i>	
10.	R1-10	AMX, P, CXM, NX, S, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
11.	R1-11	AMX, P, CZ, NX, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1, int2</i>
12.	R1-12	P, CZ, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1, int2</i>
13.	R1-13	AMX, P, CZ, S, TET, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaCTX-M2</i>	<i>int1</i>
14.	R1-14	AMX, P, CZ, S, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaCTX-M1</i>	<i>int1</i>
15.	R1-15	AMX, P, CD, COT	+	<i>sul1, sul2</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1</i>
16.	R1-16	P, CZ, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i>	<i>int1, int2</i>
17.	R1-17	AMX, P, CD, COT, TR	+	<i>dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i>	<i>int1</i>
18.	R1-18	AMX, P, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1</i>
19.	R1-19	AMX, P, CZ, CD, COT, TR	+	<i>sul1, dfrA1</i>	<i>blaTEM</i>	<i>int1, int2</i>
20.	R1-20	P, CZ, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1, int2</i>
21.	R1-21	AMX, P, CZ, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
22.	R1-22	AMX, P, CZ, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
23.	R1-23	AMX, P, CZ, E, CD, COT, TR	+	<i>sul1, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
24.	R1-24	AMX, P, CZ, CXM, CD	+		<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	
25.	R1-25	AMX,CZ, OF, S, CD	+		<i>blaTEM</i>	
26.	R1-26	AMX, P, CZ, GEN, CD	+		<i>blaTEM</i>	
27.	R1-27	P, NX, S, CD, COT	+		<i>blaTEM</i> + <i>blaCTX-M1</i>	<i>int1</i>
28.	R1-28	P, CZ, CXM, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
29.	R1-29	AMX, P, CZ, CXM, NX, OF, S, TET, E, CD, COT, TR	—	<i>sul1, sul2, dfrA1</i>		<i>int1</i>
30.	R1-30	AMX, P, CZ, CXM, NX, S, TET, E, CD, COT, TR	—	<i>sul1, sul2, dfrA1, tet(A)</i>		<i>int1</i>
31.	R1-31	P, NX, TET, S, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
32.	R1-32	AMX, P, CZ, CXM, S, TET, E, CD, COT, TR	+	<i>dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
33.	R1-33	AMX, NX, GEN, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaCTX-M1</i> + <i>blaCTX-M9</i>	<i>int1</i>
34.	R1-34	P, CZ, CXM, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaCTX-M1</i>	<i>int1</i>
35.	R1-35	AMX, P,CXM, TET	+	<i>tet(A)</i>	<i>blaTEM</i>	
36.	R2-1	AMX, CZ, TET, CD, COT, TR	+	<i>sul1, sul2, dfrA1, tet(A)</i>	<i>blaTEM</i>	<i>int1</i>
37.	R2-2	AMX, P, CXM, S, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
38.	R2-3	P, CZ, S, CD	+		<i>blaTEM</i> + <i>blaCTX-M9</i>	
39.	R2-4	P, CXM, NX, OF, E, CD, COT, TR	+	<i>dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i>	<i>int1</i>
40.	R2-5	AMX, P, CZ, CD, COT, TR	+	<i>sul1, sul2, dfrA1, dfrA12</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1</i>
41.	R2-6	AMX, P, CZ, TET, E, CD, TR	+	<i>dfrA1, tet(A)</i>	<i>blaTEM</i>	<i>int1</i>
42.	R2-7	AMX, P, CZ, CXM, E, CD	—			<i>int1</i>
43.	R2-8	AMX, P, CZ, CXM, TET, E, CD, COT	+	<i>sul1, sul2, tet(A)</i>	<i>blaTEM</i>	<i>int1</i>
44.	R2-9	AMX, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaCTX-M2</i>	<i>int1</i>
45.	R2-10	AMX, P, CZ, CXM, NX, S, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
46.	R2-11	AMX, P, CZ, CXM, S, TET, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1, tet(A)</i>	<i>blaTEM</i> + <i>blaSHV</i>	<i>int1</i>
47.	R2-12	AMX, P, CZ, NX, TET, CD, COT, TR	+	<i>sul1, sul2, dfrA1, dfrA12, tet(A)</i>	<i>blaTEM</i> + <i>blaSHV</i>	<i>int2</i>
48.	R2-13	AMX, P, CZ, CXM, NX, CD, COT, TR	+	<i>sul1, dfrA1, dfrA12</i>	<i>blaTEM</i> + <i>blaCTX-M1</i>	<i>int1</i>
49.	R2-14	AMX, P, CZ, CD	+		<i>blaTEM</i>	<i>int1</i>
50.	R2-15	AMX, P, CZ, CXM, TET, CD, COT, TR	—	<i>sul1, dfrA1, dfrA12, tet(A)</i>		<i>int1</i>
51.	R2-16	AMX, P, CZ, CXM, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1</i>
52.	R2-17	AMX, P, CZ, CXM, E, TR	—	<i>dfrA1</i>		<i>int2</i>
53.	R2-18	AMX, P, CZ, CXM, S, CD, COT	+	<i>sul1, sul2</i>	<i>blaTEM</i> + <i>blaCTX-M9</i>	
54.	R2-19	AMX, CZ, S, CD	+		<i>blaTEM</i>	
55.	R2-20	AMX, P, CZ, TET, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1, int2</i>
56.	R2-21	AMX, P, CZ, E, CD	+		<i>blaTEM</i> + <i>blaCTX-M2</i>	
57.	R2-22	AMX, P, CXM, E, CD	+		<i>blaTEM</i> + <i>blaCTX-M2</i>	
58.	R2-23	AMX, P, CZ, S, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaCTX-M1</i>	<i>int1</i>
59.	R2-24	AMX, P, CZ, E, CD, COT	+	<i>sul1, sul2</i>	<i>blaTEM</i>	
60.	R2-25	AMX, P, CZ, NX, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1</i>
61.	R2-26	AMX, P, CZ, NX, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1</i>
62.	R2-27	P, CZ, CXM, S, CD, COT	+	<i>sul1</i>	<i>blaTEM</i> + <i>blaSHV</i>	
63.	R2-28	AMX, P, CZ, CXM, NX, TET, E, CD, COT	+		<i>blaTEM</i>	
64.	R2-29	AMX, P, CZ,IPM, CD	+		<i>blaTEM</i> + <i>blaSHV</i>	
65.	R2-30	AMX, P, CZ, CXM, IPM, TR	—	<i>dfrA1, dfrA12</i>		<i>int2</i>
66.	R2-31	AMX, P, CXM, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i>	<i>int1</i>
67.	R3-1	P, E, CD	—			
68.	R3-2	P, CZ, CXM, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	<i>int1</i>
69.	R3-3	AMX, P, CZ,CXM, TET, E, COT, TR	+	<i>sul1, dfrA1, dfrA12</i>	<i>blaTEM</i>	<i>int1</i>
70.	R3-4	AMX, P, CZ,CXM	+		<i>blaTEM</i> + <i>blaSHV</i> + <i>blaOXA</i>	
71.	R3-5	AMX, P, CZ, CD	+		<i>blaTEM</i> + <i>blaSHV</i>	<i>int1</i>
72.	R3-6	AMX, P, CZ,E, CD, TR	+		<i>blaTEM</i>	

Table 4 (continued)

S. no	Isolate code	Antibiotic phenotypic resistance	ESBL phenotype	Antibiotic genes detected	Beta-lactam genes	Integrans
73.	R3-7	AMX, P, CD, COT	+		<i>bla</i> TEM	
74.	R3-8	AMX, P, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> TEM + <i>bla</i> SHV	<i>intl1</i>
75.	R3-9	AMX, P, NX, CD, TR	—	<i>dfrA1</i>		<i>intl1</i>
76.	R3-10	AMX, P, CZ, CXM, E, CD, COT, TR	—	<i>sul1, sul2, dfrA1</i>		<i>intl1</i>
77.	R3-11	AMX, P, CZ, CXM, CD, COT, TR	—	<i>sul1, sul2, dfrA1, dfrA12</i>		
78.	R3-12	AMX, P, CZ, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> TEM	<i>intl1</i>
79.	R3-13	AMX, P, CZ, CXM, CD, TR	+	<i>dfrA1</i>		<i>intl1</i>
80.	R3-14	AMX, P, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> TEM	<i>intl1</i>
81.	R3-15	AMX, P, CZ, CD, COT	+	<i>sul1, sul2</i>	<i>bla</i> TEM	<i>intl1</i>
82.	R3-16	NX, GEN, E, CD	+			<i>intl1</i>
83.	R3-17	P, E, CD, COT	+	<i>sul1, sul2</i>	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> CTX-M2	
84.	R3-18	AMX, CZ, CXM, NX, OF, IPM, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> CTX-M9	<i>intl1</i>
85.	R3-19	AMX, P, CZ, CXM, NX, TET, CD, COT	+	<i>sul1, sul2, tet(A)</i>	<i>bla</i> SHV + <i>bla</i> CTX-M1	
86.	R3-20	AMX, P, CZ, TET, CD	+	<i>tet(A)</i>	<i>bla</i> TEM	
87.	R3-21	AMX, P, CZ, CXM, E, CD, COT	+	<i>sul1, sul2</i>	<i>bla</i> TEM	
88.	R3-22	AMX, P, CZ, CXM, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> TEM	<i>intl1</i>
89.	R3-23	AMX, P, CZ, CXM, NX, OF, GEN, S, IPM, TET, COT, TR	—	<i>sul1, sul2, dfrA1, dfrA12, tet(A)</i>		<i>intl1</i>
90.	R3-24	AMX, P, CZ, CXM, CD, TR	+		<i>bla</i> TEM	
91.	R3-25	AMX, P, CZ, CXM, TET, CD	+	<i>tet(A)</i>	<i>bla</i> TEM	
92.	R3-26	AMX, P, CZ, S, TET, CD, COT	+	<i>sul1</i>	<i>bla</i> TEM	
93.	R3-27	AMX, P, CZ, CXM, TET, E, CD, COT, TR	+	<i>sul1, sul2</i>	<i>bla</i> TEM	
94.	R3-28	AMX, P, CZ, CXM, S, CD, TR	+		<i>bla</i> TEM	
95.	R3-29	AMX, P, CZ, CD, COT, TR	+		<i>bla</i> TEM	
96.	R3-30	AMX, P, CZ, CXM, NX, OF, GEN, S, IPM, TET, CD, COT	—	<i>sul1, sul2, tet(A)</i>		<i>intl1</i>
97.	R4-1	AMX, P, CZ, CD, TR	+	<i>dfrA1</i>		<i>intl1</i>
98.	R4-2	P, CZ, CXM, COT	+	<i>sul1, sul2</i>	<i>bla</i> TEM	
99.	R4-3	AMX, P, CZ, CD	+		<i>bla</i> TEM	
100.	R4-4	AMX, P, CZ, CXM, S, E, CD	+		<i>bla</i> TEM	
101.	R4-5	AMX, P, CZ, CD, COT, TR	+	<i>sul2</i>	<i>bla</i> TEM	
102.	R4-6	AMX, P, CD, COT	+	<i>sul1, sul2</i>	<i>bla</i> TEM	
103.	R4-7	AMX, P, CZ, CXM, NX, E, CD	+		<i>bla</i> TEM	
104.	R4-8	AMX, P, CZ, CXM, S, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> TEM	<i>intl1</i>
105.	R4-9	AMX, P, CZ, CD, COT	+	<i>sul1</i>	<i>bla</i> TEM + <i>bla</i> SHV + <i>bla</i> OXA	
106.	R4-10	AMX, P, CZ, CXM, NX, GEN, S, IPM, TET, E, CD, TR	—	<i>dfrA1</i>		<i>intl1</i>
107.	R4-11	AMX, P, CZ, CXM, TET, E, CD, COT, TR	—	<i>dfrA1</i>		<i>intl1</i>
108.	R4-12	AMX, P, CZ, CXM, NX, TET, E, CD, COT	—	<i>tet(A)</i>		<i>intl1</i>
109.	R4-13	AMX, P, CZ, CXM, NX, OF, GEN, S, IPM, TET, E, CD, COT, TR	—	<i>sul1, dfrA1, tet(A)</i>		<i>intl1</i>
110.	R4-14	AMX, P, CZ, CXM, NX, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> TEM	<i>intl1</i>
111.	R4-15	AMX, P, CZ, CXM, S, E, CD, COT	+		<i>bla</i> TEM	
112.	R4-16	AMX, P, CZ, GEN, S, E, CD, COT, TR	+	<i>sul1, sul2, dfrA1</i>	<i>bla</i> TEM	<i>intl1</i>
113.	R4-17	AMX, P, CZ, CD	+		<i>bla</i> TEM	
114.	R4-18	AMX, P, CZ, CD	+		<i>bla</i> TEM	
115.	R4-19	AMX, P, CZ, CXM, NX, GEN, S, CD, COT	+	<i>sul1, sul2</i>	<i>bla</i> TEM	
116.	R4-20	AMX, P, CZ, S, TET, CD	+		<i>bla</i> TEM	
117.	R4-21	AMX, P, CZ, NX, S	+		<i>bla</i> TEM	
118.	R4-22	AMX, P, CZ, CXM, TET, E, CD, COT	—	<i>sul1, sul2</i>		
119.	R4-23	AMX, P, CZ, CXM, NX, GEN, TET, E, CD, TR	—	<i>dfrA1</i>		<i>intl1</i>
120.	R4-24	AMX, P, CZ, CD	+		<i>bla</i> TEM	
121.	R4-25	AMX, P, CZ, CD, TR	+	<i>dfrA1</i>	<i>bla</i> TEM	<i>intl1</i>

+: positive; —: negative; sulfonamide genes (*sul1, sul2*); trimethoprim genes (*dfrA1, dfrA12*); tetracycline genes (*tetA*).

(Fig. 2). High percentage of *intl1* gene agrees with the earlier studies that have proved the omnipresent prevalence of class 1 integron as a result of the high incidence of anthropopressure and its pivotal role in spreading the MDR trait through horizontal gene transfer (Cambray et al., 2010; Domingues et al., 2012; Makowska et al., 2020) while the second class of integrons (*intl2*) is less common than *intl1* (Chigor et al., 2020; Obayiuwana et al., 2018; Koczura et al., 2014; Kaushik et al., 2018; Kotlarska et al., 2015). Initially found in the clinical strains, class 1 integrons are now widespread in environmental bacteria (Lupo et al., 2012; Gillings, 2014; Gillings et al., 2015; Makowska et al., 2020). Additionally, class 1 integrons are a well-known proxy for anthropogenic pollution and their property to acquire a diverse array of gene cassettes (Wright et al., 2008; Gillings et al., 2015) possessing combinations of antibiotic resistance genes (Ma et al., 2017) thereby turning a susceptible bacteria (initially resistant to 1 antibiotic) into a resistant one. Integrons that significantly contribute to the spread of resistant factors have been previously detected in various rivers and pristine environments worldwide (Su et al., 2012; Makowska et al., 2020).

3.5. Determination of ARGs and its association with integrons

The genes for sulfonamides resistance (*sul* and *dfr*) are genetically linked to class 1 integrons and the isolates positive for integrons also displayed resistance towards antibiotics belonging to class-sulfonamide (co-trimoxazole and trimethoprim), the detection of *sul* genes (*sul1* and *sul2*) and *dfr* genes (*dfrA1* and *dfrA12*) was carried out using PCR. Out of 73 integron positive strains, the *sul1* gene was detected in 53 (72.60%) isolates (Fig. 3, Table 4). Some of the isolates that showed negative results for the *intl1* gene were also found to harbour the *sul1* gene (10; 8.26%). Tetracycline resistance gene (*tetA*) has also been reported in aquatic environments (Zhai et al., 2016) but in the present study the gene was found in very few isolates.

A significant correlation exists and has been reported between class I integrons and sulfonamide resistance genes (Dungan et al., 2018; Nolvak et al., 2016; Su et al., 2012; Ma et al., 2017; Uyaguari-Díaz et al., 2018) that may be attributed to the presence of *sul1* gene in the conserved segment of class 1 integron (Dungan et al., 2018; Makowska et al., 2016). The present study also reported a strong correlation between *sul1*-*intl1* ($R = 0.96$)

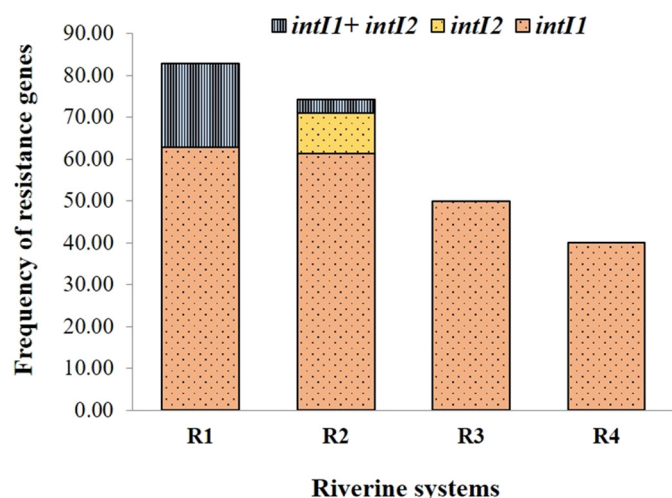


Fig. 2. Prevalence of integrase genes (%) (*intI1* and *intI2*) in (R1-Ganga, R2-Gomti, R3-Narmada, R4-Kaveri) riverine systems.

and *sul2-intI1* ($R = 0.97$) gene. Jia et al. (2018) also reported in a study the correlation between *sul1-intI1* was ($R = 0.96$) in Ba river China (Jia et al., 2018). The presence of the *sul1* gene in the integron positive isolates is justified by the structural make-up of the integron whereas the existence of *sul2* and *intI1* genes together may be due to the distribution of the genes in larger mobile elements such as plasmids (Labar et al., 2012). The aquatic environment further provides a vital platform for the easy dispersal and evolution of ARB and ARGs with such mobile genetic elements.

A total of 48 (65.75%) isolates were found to carry both *sul1* and *sul2* genes. A similar distribution of *sul* genes in river water have been reported earlier (Jiang et al., 2013; Makowska et al., 2016). On the contrary, in some studies, *sul1* gene is more abundant than *sul2* (Gao et al., 2012). The high percentage of *sul* genes may be associated with high concentrations of sulfonamides in the environment (Khan et al., 2013) as these are one of the most frequently used antibiotics by humans (Nowrotek et al., 2017). The study also displays a positive correlation between sulfonamide and *sul* genes ($sul1R = 0.9$, $sul2R = 0.8$) (Fig. 4). Tuo et al. (2018) reported in Furan River the prevalence of colistin resistance strains were found in higher number. They isolated total 18 *mcr-1* positive isolates, out of which *E. coli* (17), *Enterobacter cloacae* (01) and 06 *mcr-3*-positive isolates (04 *Aeromonas hydrophila* and *Aeromonas veronii*) were found. In further study of antibiotic resistance

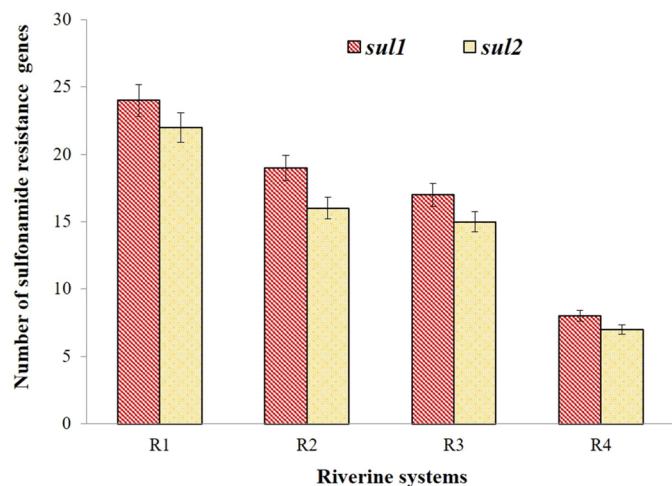


Fig. 3. Sulfonamide resistant genes *sul1* and *sul2* (R1-Ganga, R2-Gomti, R3-Narmada, R4-Kaveri) in rivers.

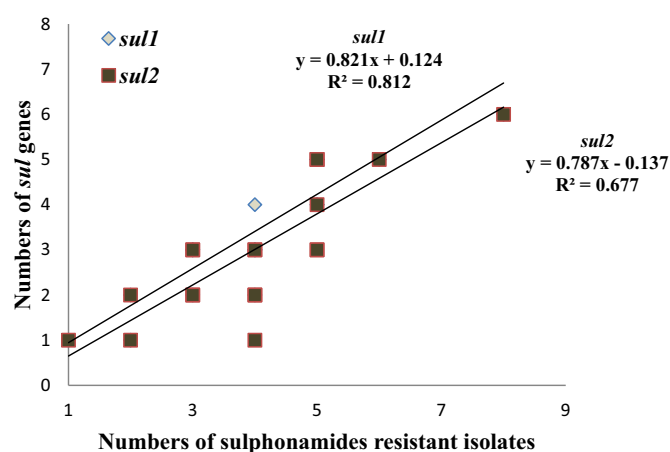


Fig. 4. Correlation analysis between the numbers of sulfonamides resistant isolates and detection of *sul* genes (*sul1*, *sul2*) in riverine system.

in Furan River 15 ARGs and class 1 integrons genes (*intI1*) were found. 09 genes *sul1*, *sul2*, *intI1*, *aac(60)-Ib-cr*, *blaCTX-M*, *tetM*, *ermB*, *qnrS*, and *aph(30)-IIIa* were observed in higher concentration between 70 and 100% in samples. Besides, in riverine system rarely detected genes like *mcr-1*, *blaKPC*, *blaNDM* and *vanA* were also observed. Furthermore, genes encoding resistance to trimethoprim *dfrA1* and *dfrA12* were also detected in integron bearing isolates. Out of 73 integrons positive, 63 (86.30%) isolates showed resistance for the trimethoprim. Upon amplification 64 (87.67%) isolates were found carry *dfrA1* gene with 392 bp long product whereas 9 (40.90%) isolates contain 319 bp corresponding to *dfrA12*. Trimethoprim resistance genes (*dfrA1* and *dfrA12*) were also detected in integron-positive strains isolated from the Yangtze River basin in China (Chen et al., 2010). As trimethoprim resistance genes (*dfrA1*, *dfrA12*) are part of the gene cassettes elements (Canal et al., 2016; Kaushik et al., 2018), the presence of *dfr* genes in integron positive isolates supports the study (Table 4).

3.6. Correlations analysis among physicochemical properties, sulfonamide resistance genes (*sul1*, *sul2*) and integrons (*intI1*, *intI2*)

The correlation among the physicochemical properties of water, abundance of sulfonamide genes (*sul1*, *sul2*) and integrons (*intI1*, *intI2*) were evaluated by the Pearson's correlation coefficients. DO showed negative correlation with other physicochemical parameters, sulfonamide resistance genes (*sul1*, *sul2*) and integrons (*intI1*, *intI2*). Temperature, EC, TDS and pH displayed a positive correlation whereas COD and BOD were negatively correlated with sulfonamide resistance genes and integrons. Significant correlations existed between sulfonamide resistance genes (*sul1*, *sul2*) and integrons (*intI1*, *intI2*) thereby indicating integrons as typical factors responsible for residing the ARGs and hence its distribution and dissemination between different bacterial species (Table 5). The present results and several previous studies have consistently proved this finding (Zheng et al., 2017; Zhou et al., 2017; Lv et al., 2020).

3.7. Characterization of beta-lactamase genes

Multiplex PCR was performed to spot the presence of *bla* genes in isolates that were found positive in the Double Disc Synergy (DDST) test. A total of 103 (85.15%) isolates were found ESBL positive among 121 MDR positive isolates. Out of 103 isolates, ESBL genes were detected in 100 (97.08%) isolates. *blaTEM* (93.0%) was the most prevalent ESBL gene that was found alone as well as in combination with other *bla* genes (Fig. 5). The wide distribution of *blaTEM* gene has also been reported in water samples collected from river Ganga (Singh et al., 2016) and Yamuna (Bajaj et al., 2015). The wide occurrence of *blaTEM*

Table 5Correlation between physico-chemical parameters, sulfonamide genes *sul1*, *sul2* and integrons *int11*, *int12*.

Correlation coefficient	pH	DO	Temp	EC	BOD	COD	TDS	<i>sul1</i>	<i>sul2</i>	<i>int11</i>	<i>int12</i>
pH	1.000										
DO	−0.890	1.000									
Temp	0.733	−0.876	1.000								
EC	0.648	−0.902	0.951	1.000							
BOD	−0.012	−0.400	0.225	0.516	1.000						
COD	−0.087	−0.345	0.213	0.503	0.995	1.000					
TDS	0.548	−0.702	0.958	0.875	0.111	0.127	1.000				
<i>sul1</i>	0.503	−0.438	0.766	0.542	−0.405	−0.386	0.864	1.000			
<i>sul2</i>	0.475	−0.366	0.698	0.451	−0.501	−0.482	0.805	0.994	1.000		
<i>int11</i>	0.612	−0.428	0.676	0.414	−0.566	−0.569	0.734	0.960	0.975	1.000	
<i>int12</i>	0.832	−0.626	0.721	0.493	−0.452	−0.488	0.682	0.852	0.858	0.944	1.000

gene in the ESBL⁺ isolates is mainly due to the horizontal gene transfer through plasmids. The combinations that were observed include: *blaTEM* + *blaSHV* + *blaOXA* = 11.65%, *blaTEM* + *blaSHV* = 12.62%, and *blaTEM* + *blaCTX-M* = 9.71%. The observed ESBL-genotypes presented variation in the number of antibiotics the isolates were resistant to, with highest level of multi-drug resistance among *blaTEM* (Fig. 6). Among ESBL producing isolates 58.0% were integron positive. Chen et al. (2010) reported in a study that 65.2% ESBL producing isolates carried class1 integrons (Chen et al., 2010). In some studies ESBL producing strains with a high frequency of integrons have been reported (Machado et al., 2007). Tuo et al. (2018) reported in study *blaSHV* (4.2%), *blaTEM* (16.7%), and *blaCTX-M-9* (20.8%) genes were detected in Funan river China.

4. Conclusions

Antibiotic resistance is an early warning threat to global crisis that requires appropriate action from the public health interventions to avoid any disease outbreaks. Urbanization and industrialization have accelerated the deterioration of environment which adversely affects human health. Irrespective of different ecological niche and geo-climatic zone, we detected integrons responsible for harbouring and dissemination of resistance in riverine system. Most of the isolates conferred resistance to beta-lactams, lincosamide and sulfonamide. Resistance to trimethoprim and tetracycline was also present in the isolates from riverine systems in different climatic zones. The hydro-geological chemistry of water like presence of heavy metals and ions may affect the dissemination of resistant genes possibly through horizontal gene transfer. The abundance of *int11* and *int12* reflect the exposures of

unregulated anthropogenic activities in the riverine system. Besides, *int11* and *int12* were found to be significantly correlated with *sul1* and *sul2* genes; to be responsible for the proliferation and expansion of resistant genes. The spotting of such mobile genetic elements may provide a deeper insight into the extent of the spread of antibiotic resistance in the aquatic environment. The detection of threatening strains in the river generates alarming urgency for setting up of measures for public awareness and regulation of guidelines to look-out against the prevalent resistance and surveillance of antibiotic resistance and policies to curtail its unregulated use. Hence, a mandatory and strict policy should be adopted for the adequate treatment and reuse of polluted water to avoid, dissemination of resistant isolates into the food chain ecosystem for assisting risk management in environment.

CRedit authorship contribution statement

- Preet Chaturvedi (PC):** Methodology, Experimentation work, Writing, Resources Conceptualization, Investigation, Methodology, Writing - Original draft preparation, Project administration, Funding acquisition.
- Pankaj Chowdhary (PC) and Anuradha Singh (AS):** Sampling, Experimentation work (AMR screening), Methodology, Writing - original draft preparation

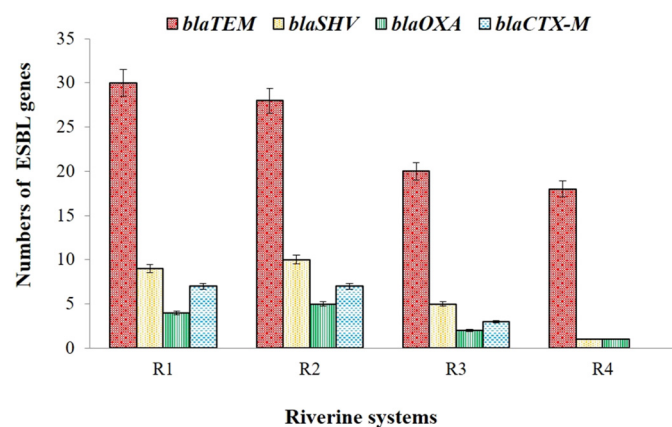


Fig. 5. Profiling of ESBL genes (*blaTEM*, *blaSHV*, *blaOXA* and *blaCTX-M*) from environment isolates from Ganga (R1), Gomti (R2), Narmada (R3) and Kaveri (R4).

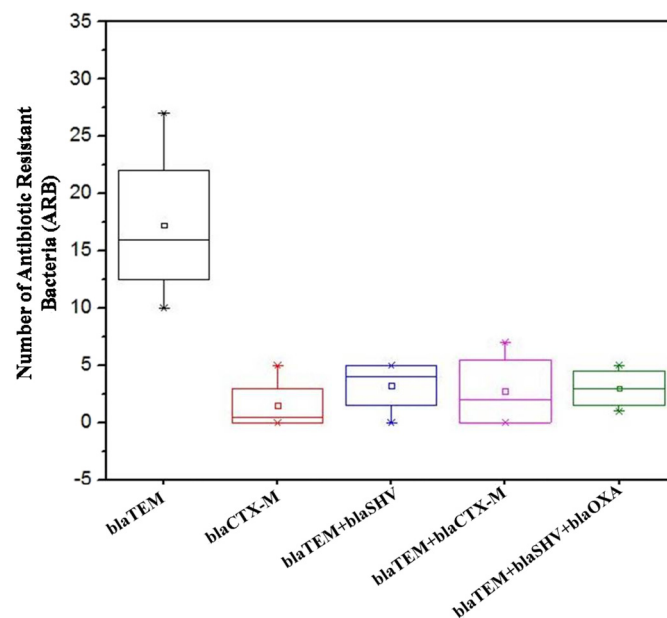


Fig. 6. Box plot showing presence of antimicrobial resistance (ARB) versus different ESBL genotypes in isolates. Solid horizontal line represents the median, the stems show the minimum and maximum values, and the cross indicates an outlier.

3. **Ashok Pandey (AP)**, and **Pratima Gupta (PG)**: Supervision, Formal analysis, Writing - reviewing and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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